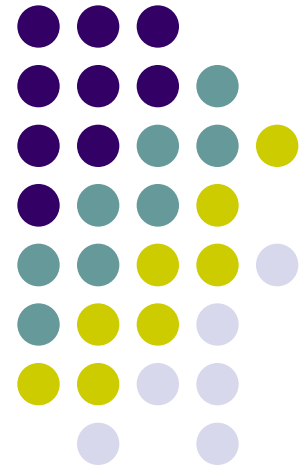
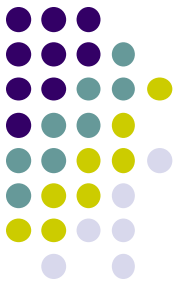


Support Models & ethics

Dynamic Modelling for
Human-centered Systems

Chapter 14





Overview of today's lecture

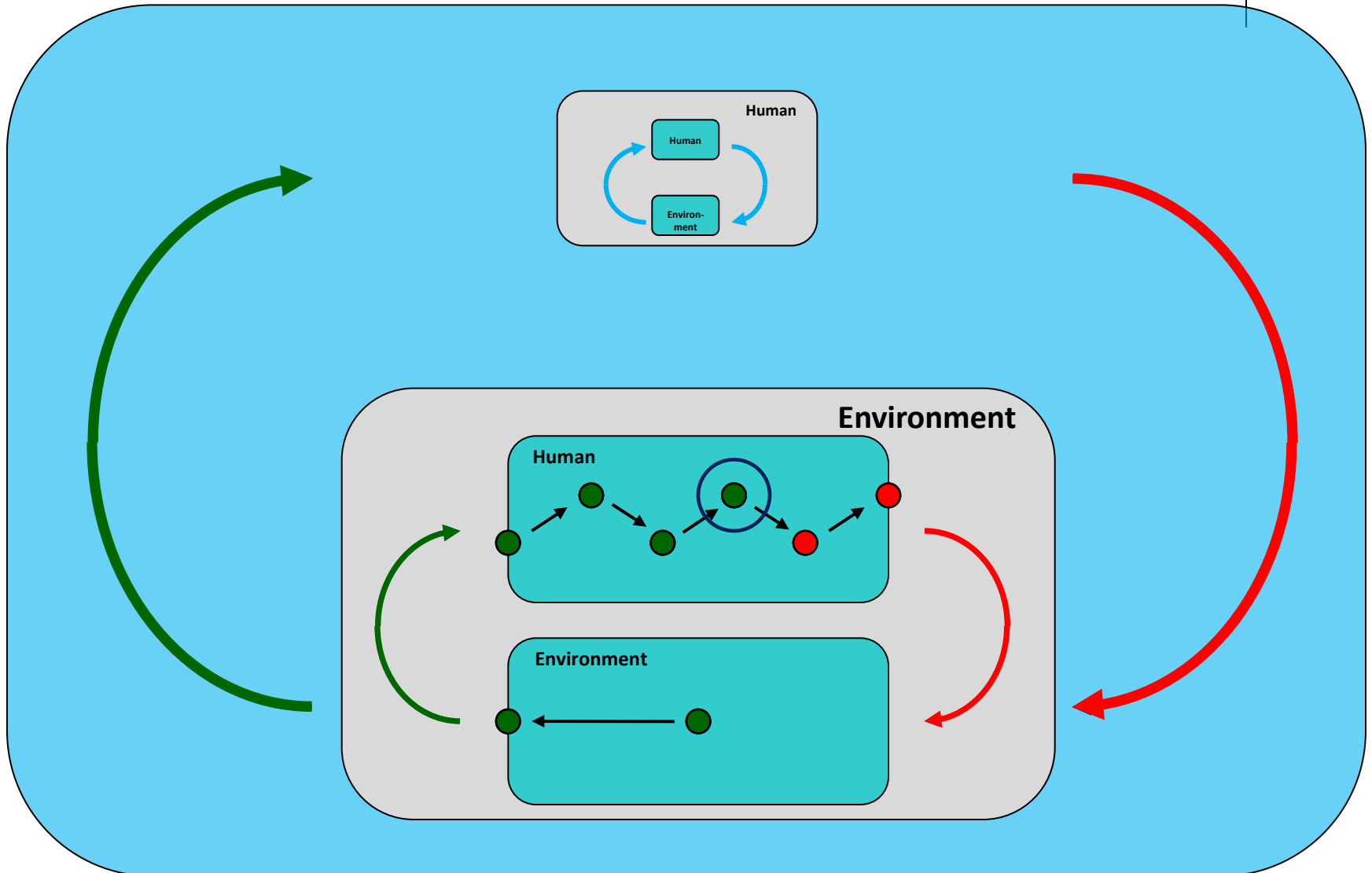
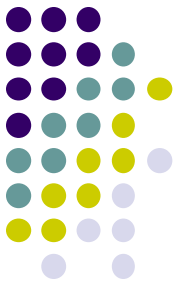
1. Summary previous lecture

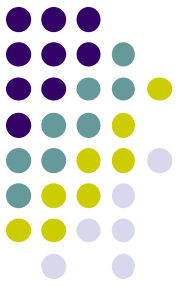
- analysis models

2. Support models

3. Ethical considerations

Using Models within AI systems





Embedding domain models

- Two roles: analysis model & support model
- Analysis model
 - Main purpose:
 - find out whether some **undesired state** occurs
 - Main requirements:
 - a domain model
 - sensor information

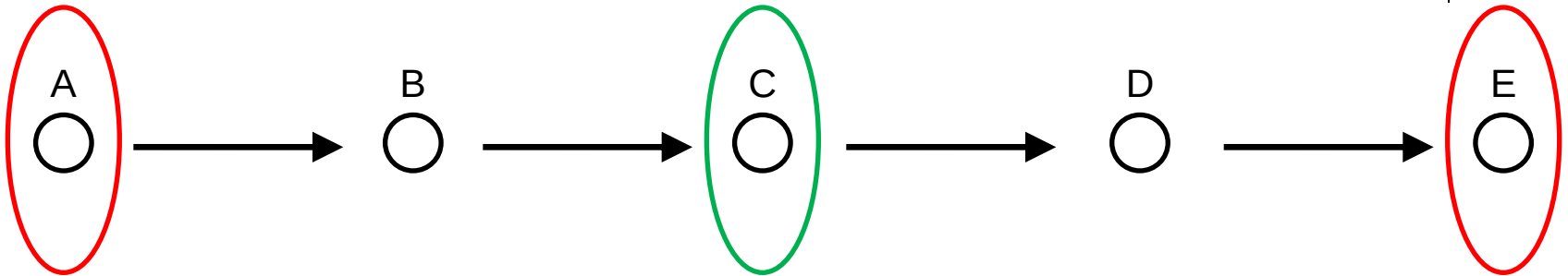
Domain Model → Analysis Model



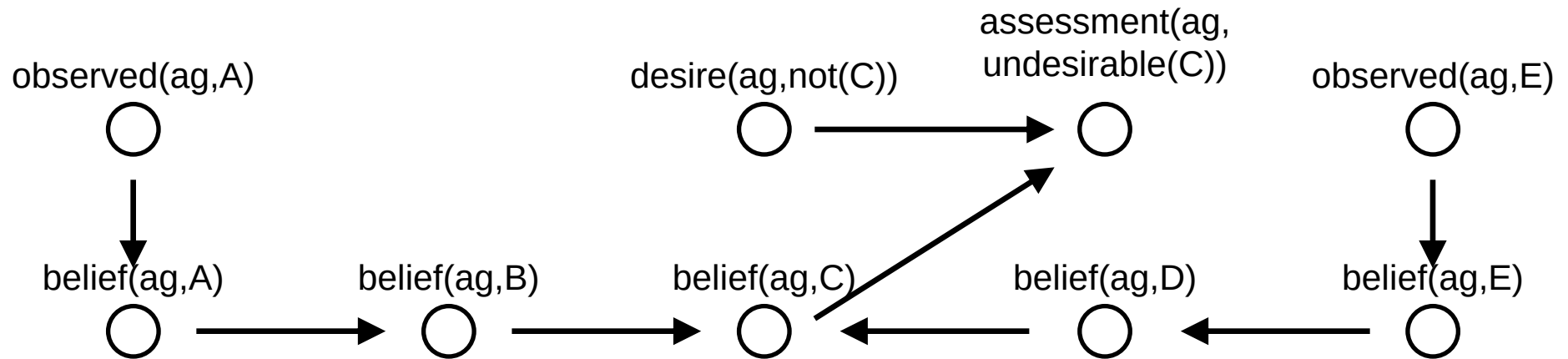
1. Decide which concepts of domain model can be **observed**
2. Introduce **observations** for these concepts
3. Decide which concept of domain model is (not) **desired**
4. Introduce a **desire** for this concept
5. Introduce **beliefs** for all concepts
6. Draw **arrows** from observations via beliefs to the desire (based on forward and/or backward reasoning)
7. Introduce an **assessment** for the concept from step 3
8. Draw **arrows** from the desire+belief to the assessment



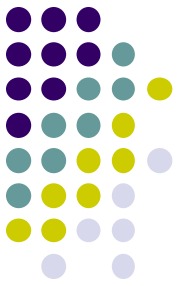
Domain Model → Analysis Model



DOMAIN

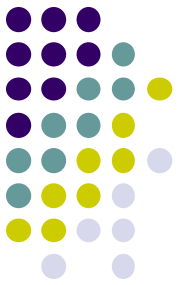


ANALYSIS



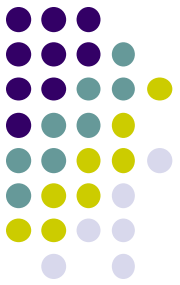
Overview of today's lecture

1. Summary previous lecture
2. **Support models**
3. Ethical considerations



Support models

- Main purpose:
 - propose a **support action** that will have a positive effect (on the undesirable state)
 - represented by a **proposed** predicate
 - in time: after having done assessment with analysis model
- Main requirements:
 - the domain and analysis model
 - a **set** of possible **support actions**

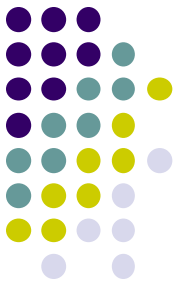


Reasoning in support models

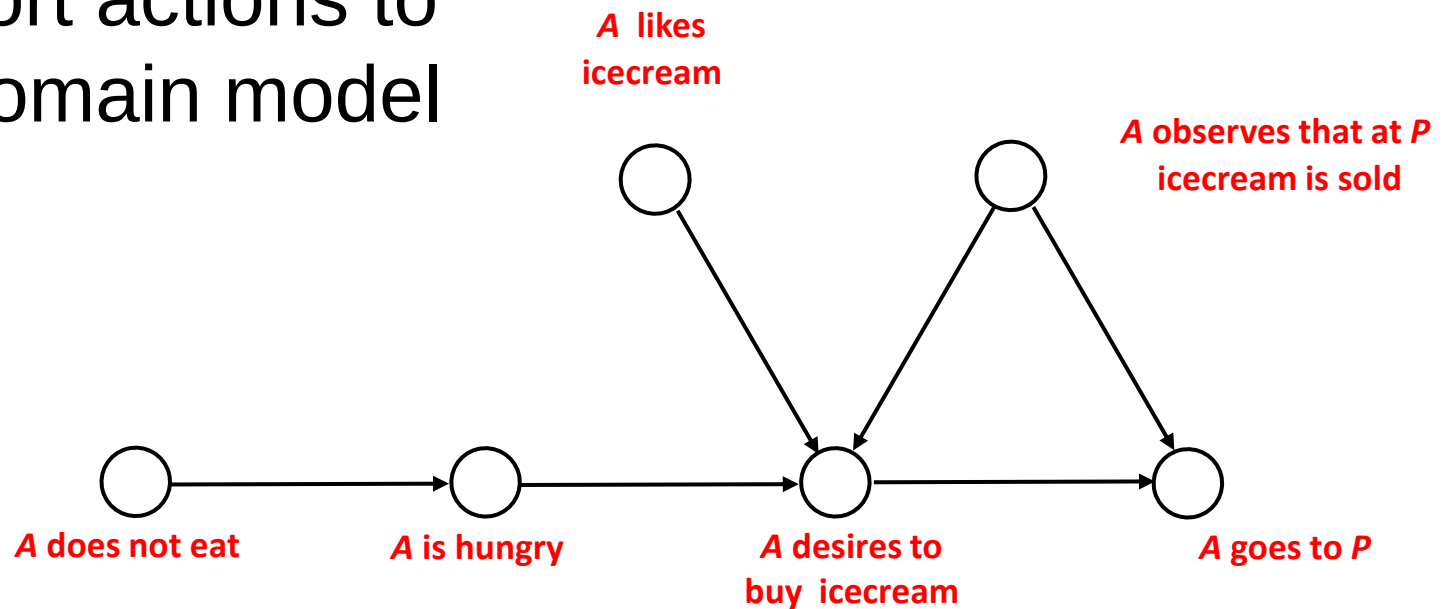
- Reasoning can be done in two ways:
 - Method 1: **backward** reasoning
 - deduct which action influences the state we are interested in
 - starting from desired state, towards actions
 - Method 2: **forward** reasoning
 - simulate effect of each possible action
 - starting from action(s), towards desired state



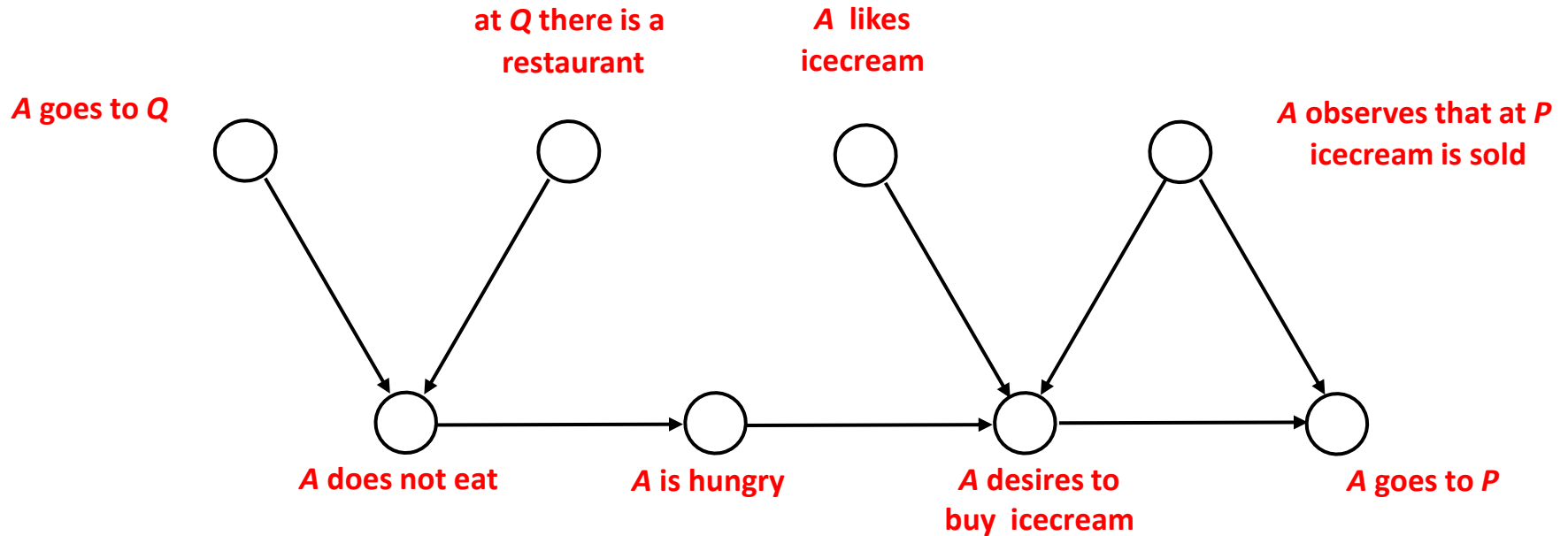
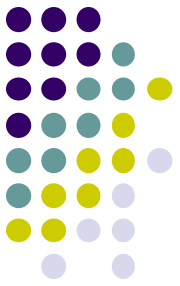
Dynamic relationships in the domain model

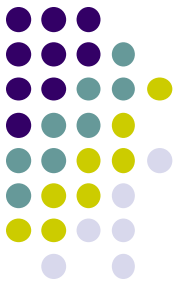


- Step 0: add possible support actions to the domain model



Extension of domain model with support actions





Extension of domain model

DDR4 From going to a restaurant to eating there

If A goes to a place Q
and at Q there is a restaurant
then A will eat

performed(A, goto(Q)) &
restaurant_at(Q)
->> performed(A, eat)

DDR5 Not becoming hungry by eating

If A does eat
then A will not be hungry

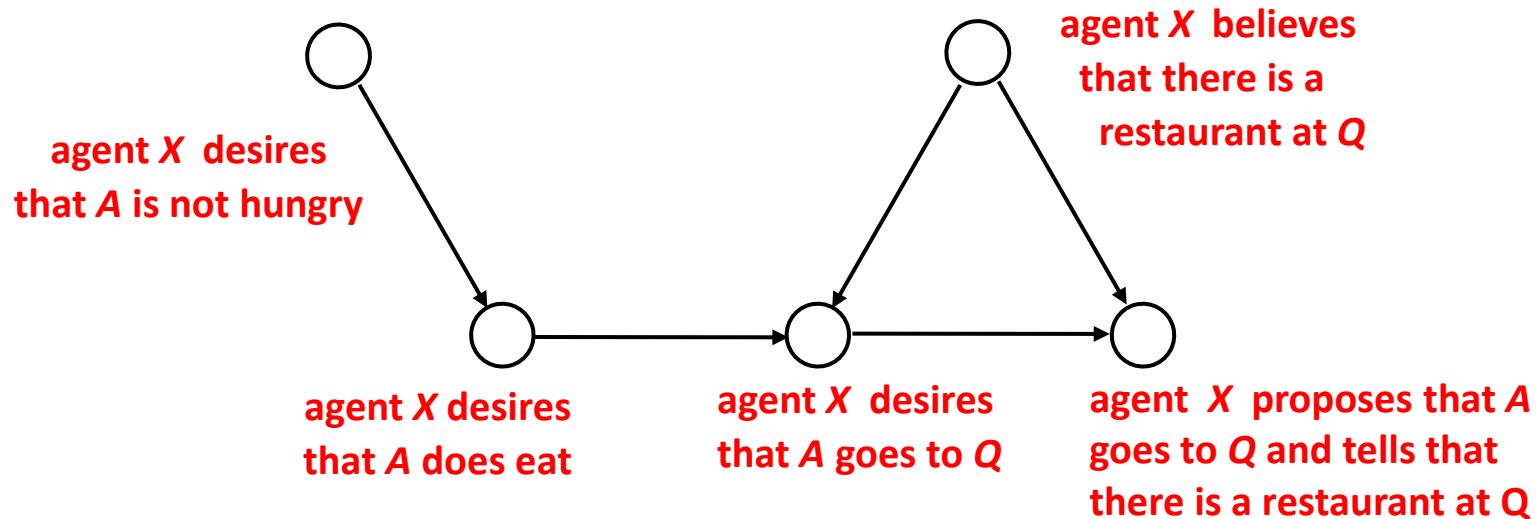
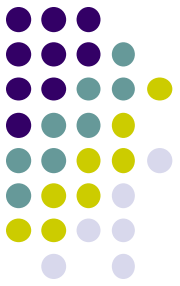
performed(A, eat) ->> not hungry(A)



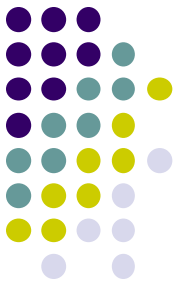
Method 1: backwards reasoning

- Approach: **deduct** which action influences the state we are interested in.
- Concepts
 - ambient agent **X** desires that **A** is not be hungry
 - ambient agent **X** desires that **A** does eat
 - ambient agent **X** desires that **A** goes to **Q**
 - ambient agent **X** believes that there is a restaurant at **Q**
 - ambient agent **X** **proposes** that **A** goes to **Q**
 - ambient agent **X** tells that there is a restaurant at **Q**

Dynamic relationships in the support model (backward)

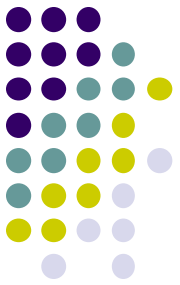


Detailed support model: concepts



<i>concept</i>	<i>formalisation</i>
ambient agent X believes that A goes to Q	belief(X: AGENT, performed(A: AGENT, goto(Q:POSITION)))
X believes that there is a restaurant at Q	belief(X: AGENT, restaurant_at(Q:POSITION))
X believes that A eats	belief(X: AGENT, performed(A: AGENT, eat))
X believes that A is not hungry	belief(X: AGENT, not(hungry(A: AGENT)))
X proposes A that A goes to Q	proposed(X: AGENT, A:AGENT, performed(A: AGENT, goto(Q:POSITION)))
X tells A that there is a restaurant at Q	told(X: AGENT, A:AGENT, restaurant_at(Q:POSITION))
X desires that A is not hungry	desire(X: AGENT, not(hungry(A:AGENT)))
X desires that A eats	desire(X: AGENT, performed(A:AGENT, eat))
X desires that A goes to Q	desire(X: AGENT, performed(A:AGENT, goto(Q:POSITION)))

Support model: formalization of relations (backward)



SDR5 Generating the desire that A eats

If ambient agent X desires that A is not be hungry
then ambient agent X will desire that A does eat

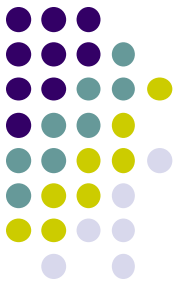
desire(X, not(hungry(A)))
->> desire(X, performed(A, eat))

SDR6 Generating the desire that A goes to Q

If ambient agent X desires that A does eat
and ambient agent X believes that at Q there is a restaurant
then ambient agent X will desire that A goes to Q

desire(X, performed(A, eat)) &
belief(X, restaurant_at(Q))
->> desire (X, performed(A, goto(Q)))

Support model: formalization of relations (backward)

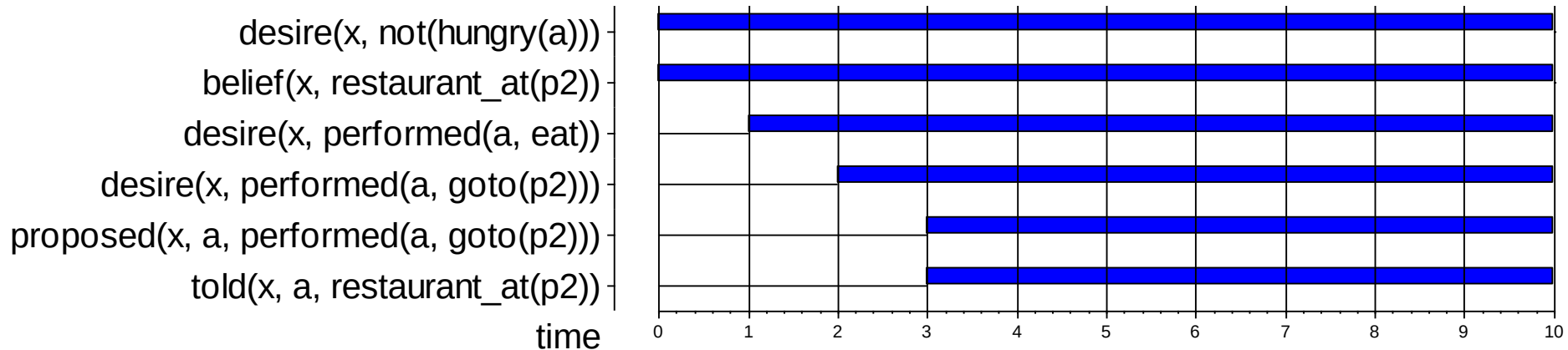
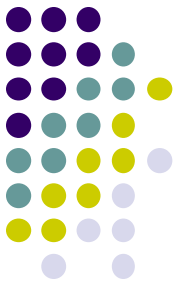


SDR7 Proposing and telling A to go to the restaurant

If ambient agent X desires that A goes to Q
and ambient agent X believes that there is a restaurant at Q
then ambient agent X will propose that A goes to Q
and ambient agent X will tell that there is a restaurant at Q

desire(X, performed(A, goto(Q))) &
belief(X, restaurant_at(Q))
->> proposed(X, A, performed(A, goto(Q))) &
 told(X, A, restaurant_at(Q))

Example trace for the backward reasoning Support model



Domain model → Support model (backward)

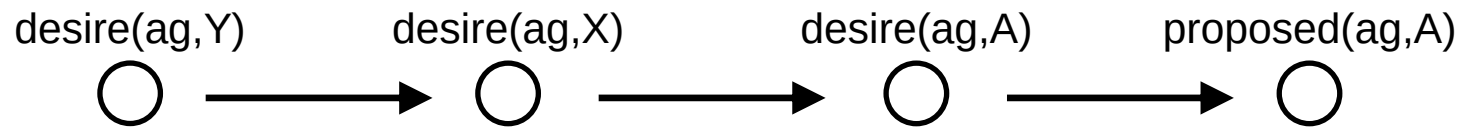
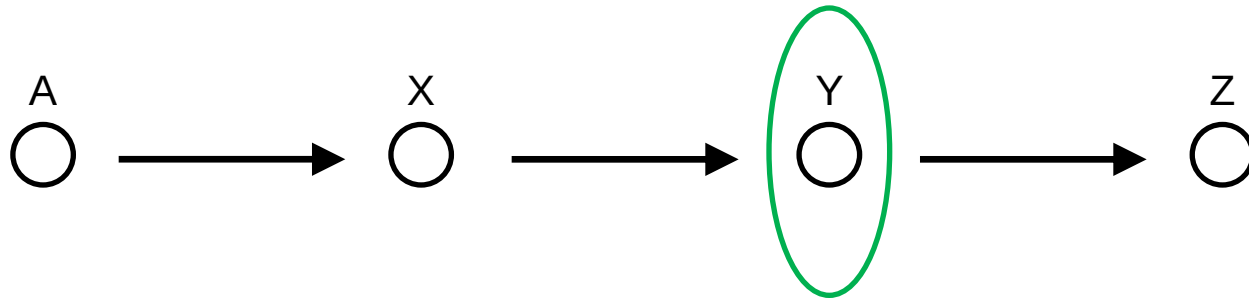
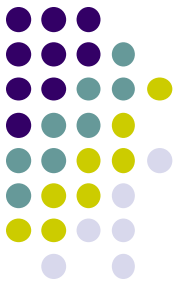


1. (Extend domain model by including **support actions**)
2. Decide which concept of domain model is (not) **desired**
3. Introduce a **desire** for this concept
4. (Propagate and remove **negative effects**)

Backward:

5. Derive **more desires** by following arrows backwards
6. If support action is reached, **propose** this action

Domain model → Support model



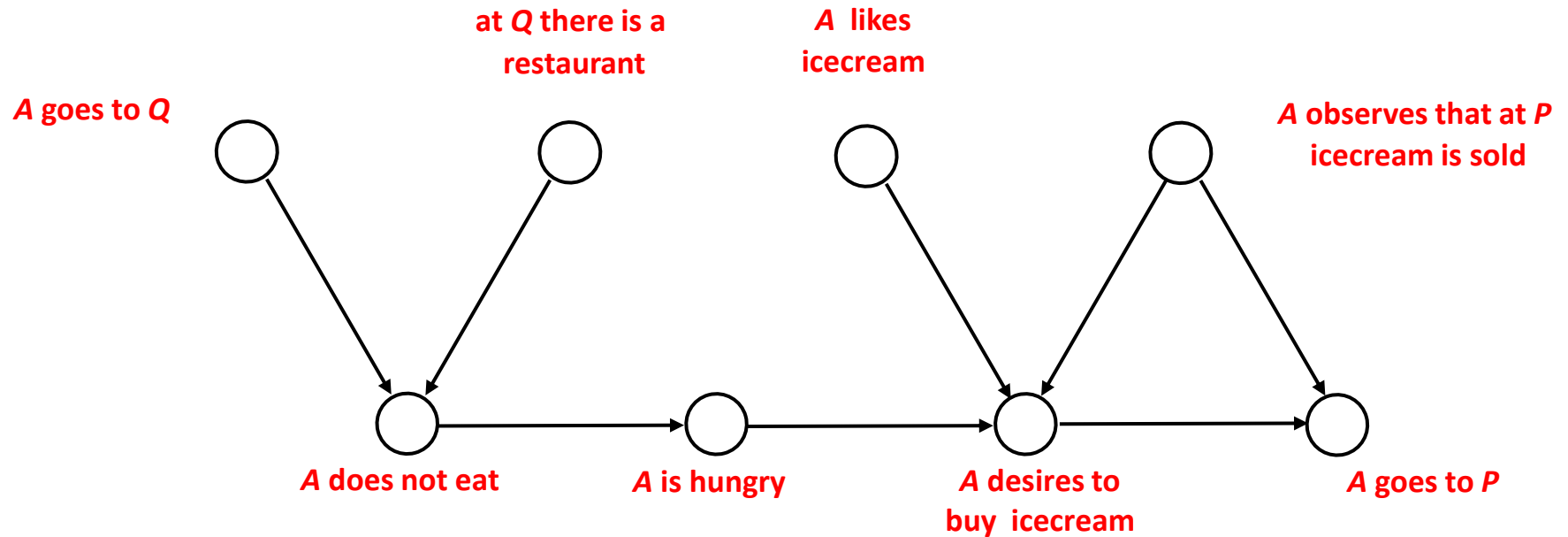
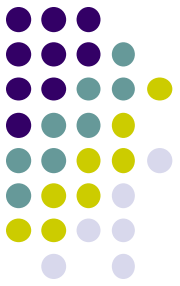
SUPPORT (backward)



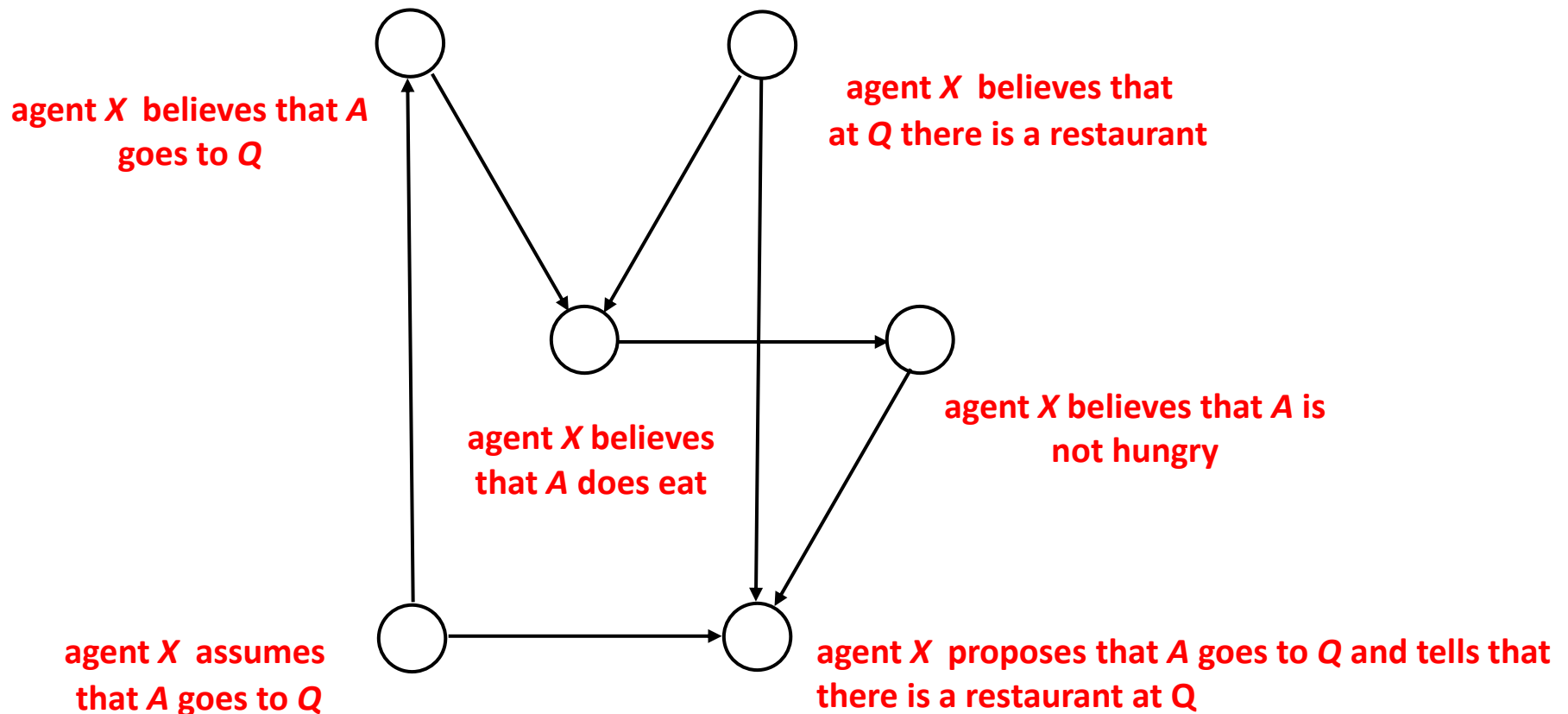
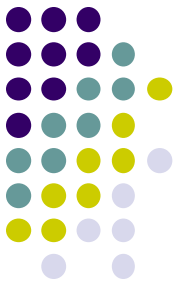
Method 2: forward reasoning

- Approach: **simulate** effect of each possible action
- Concepts:
 - ambient agent **X** **assumes** that **A** goes to **Q**
 - ambient agent **X** believes that **A** goes to **Q**
 - ambient agent **X** believes that **A** eats a meal
 - ambient agent **X** believes that at **Q** there is a restaurant
 - ambient agent **X** believes that **A** is not hungry
 - ambient agent **X** proposes that **A** goes to **Q**
 - ambient agent **X** tells that there is a restaurant at **Q**

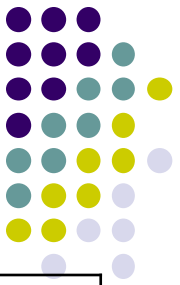
Extension of domain model with support actions



Dynamic relationships in the support model (forward)

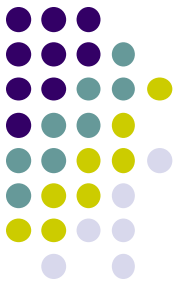


Detailed support model: concepts



<i>concept</i>	<i>formalisation</i>
ambient agent X believes that A goes to Q	belief(X: AGENT, performed(A: AGENT, goto(Q:POSITION)))
X assumes that A goes to Q	assumption(X: AGENT, performed(A: AGENT, goto(Q:POSITION)))
X believes that there is a restaurant at Q	belief(X: AGENT, restaurant_at(Q:POSITION))
X believes that A eats	belief(X: AGENT, performed(A: AGENT, eat))
X believes that A is not hungry	belief(X: AGENT, not(hungry(A: AGENT)))
X proposes A that A goes to Q	proposed(X: AGENT, A:AGENT, performed(A: AGENT, goto(Q:POSITION)))
X tells A that there is a restaurant at Q	told(X: AGENT, A:AGENT, restaurant_at(Q:POSITION))
X desires that A is not hungry	desire(X: AGENT, not(hungry(A:AGENT)))
X desires that A eats	desire(X: AGENT, performed(A:AGENT, eat))
X desires that A goes to Q	desire(X: AGENT, performed(A:AGENT, goto(Q:POSITION)))

Support model: formalization of relations (forward)



SDR1 Incorporating an assumption as a belief

If ambient agent X assumes that A goes to Q
then ambient agent X will believe that A goes to Q

`assumed( $X$ , performed( $A$ , goto( $Q$ )))`

`->> belief( $X$ , performed( $A$ , goto( $Q$ )))`

SDR2 Generating the belief that A will eat when at the restaurant

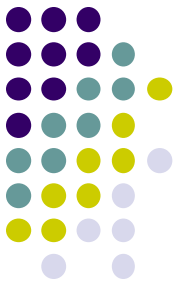
If ambient agent X believes that A goes to at place Q
and ambient agent X believes that at Q there is a restaurant
then ambient agent X will believe that A will eat

`belief( $X$ , performed( $A$ , goto( $Q$ ))) &`

`belief( $X$ , restaurant_at( $Q$ ))`

`->> belief( $X$ , performed( $A$ , eat))`

A Detailed Support Model for Reasoning Forward in Time



SDR3 Belief on not becoming hungry by eating

If ambient agent *X* believes that *A* does eat a meal
then ambient agent *X* will believe that *A* will not be hungry

belief(*X*, performed(*A*, eat))

->> belief(*X*, not(hungry(*A*)))

SDR4 Proposing and telling *A* to go to the restaurant

If ambient agent *X* assumes that *A* goes to *Q*
and ambient agent *X* believes that *A* is not hungry
and ambient agent *X* believes that at *Q* there is a restaurant
then ambient agent *X* will propose that *A* goes to *Q*
and ambient agent *X* will tell that there is a restaurant at *Q*

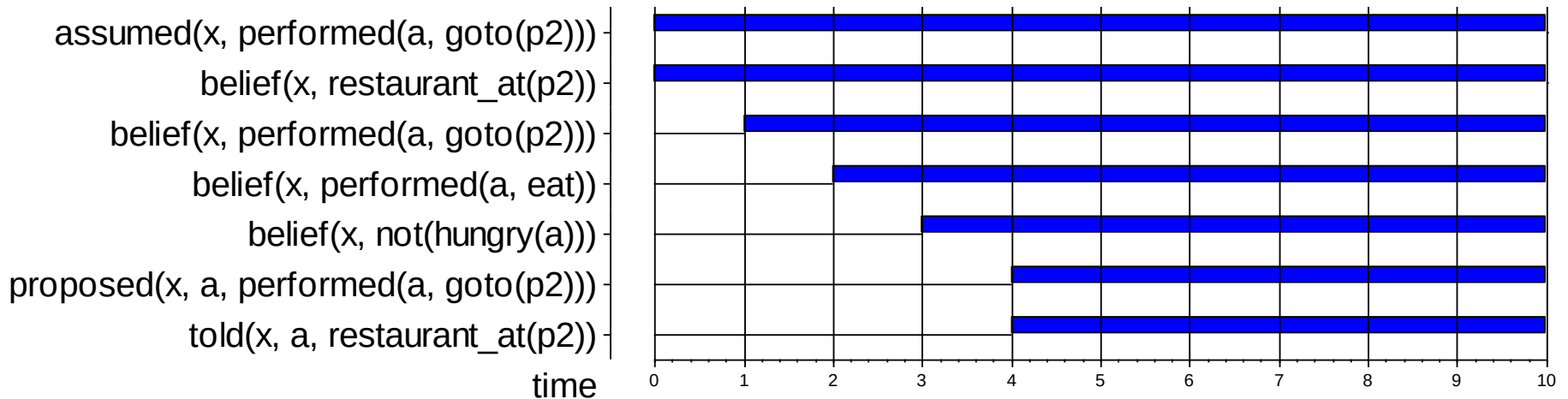
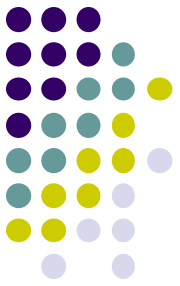
assumed(*X*, performed(*A*, goto(*Q*))) &

belief(*X*, not(hungry(*A*))) &

belief(*X*, restaurant_at(*Q*))

->> proposed(*X*, *A*, performed(*A*, goto(*Q*))) & told(*X*, *A*, restaurant_at(*Q*))

Example trace for the forward reasoning Support model



Generic process of developing support models



- Main purpose:
 - propose a **support action** that will have a positive effect (on the undesirable state)
- Main requirements:
 - the domain and analysis model
 - a set of possible **support actions**
- Two approaches
 - by deducting which action influences the state we are interested in (backward)
 - by simulation of effect of actions (forward)

Domain model → Support model (backward)



1. (Extend domain model by including **support actions**)
2. Decide which concept of domain model is (not) **desired**
3. Introduce a **desire** for this concept
4. (Propagate and remove **negative effects**)

Backward:

5. Derive **more desires** by following arrows backwards
6. If support action is reached, **propose** this action

Domain model → Support model (forward; “what if”)

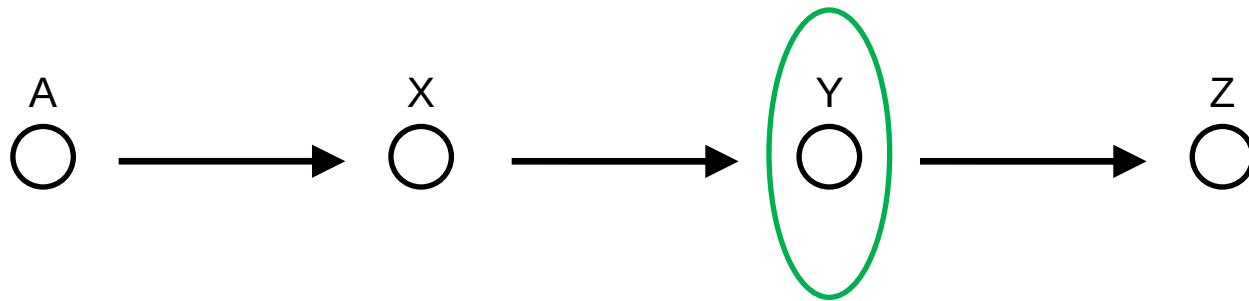
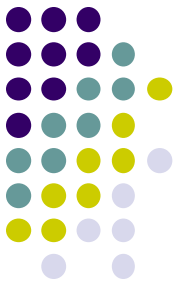


1. (Extend domain model by including **support actions**)
2. Decide which concept of domain model is (not) **desired**
3. Introduce a **desire** for this concept
4. (Propagate and remove **negative effects**)

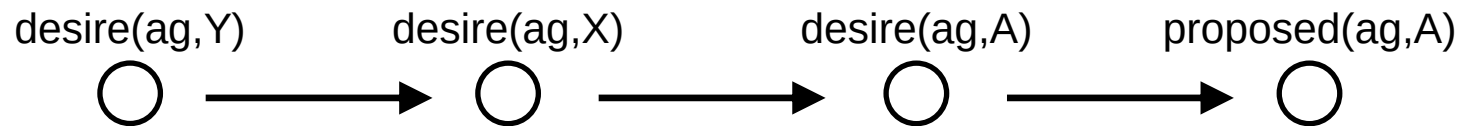
Forward:

5. Introduce an **assumption** for a selected support action
6. Derive **beliefs** by following arrows forwards
7. If desire is reached, **propose** the support action
(also include arrows from desire and assumption)

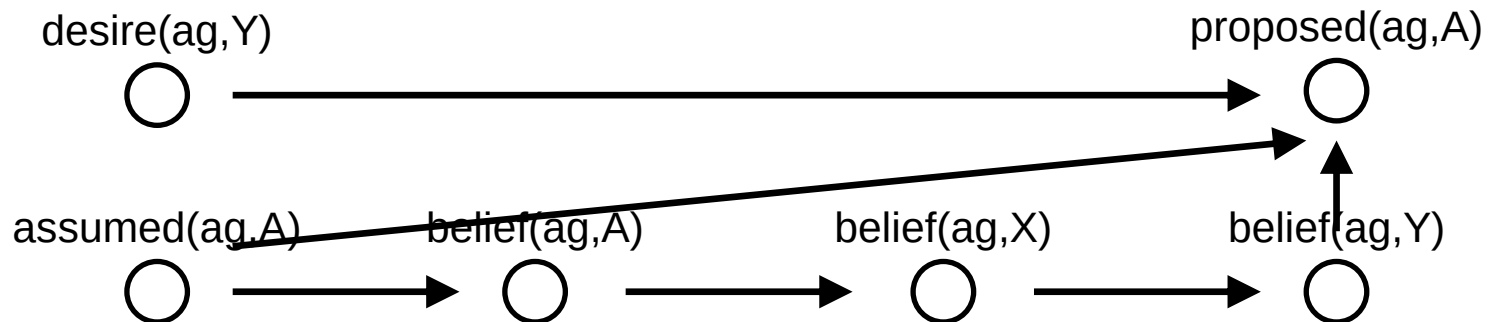
Domain model → Support model



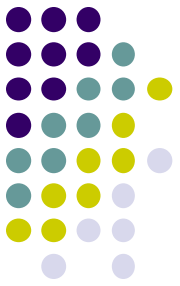
DOMAIN



SUPPORT (backward)



SUPPORT (forward)

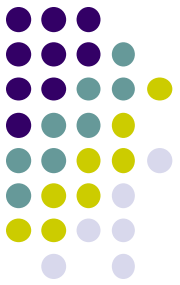


Additional Modelling Concepts

<i>concept</i>	<i>relation</i>
assumption (of a support action)	assumed : AGENT x INFO_ELEMENT
proposed support action	proposed: AGENT x INFO_ELEMENT

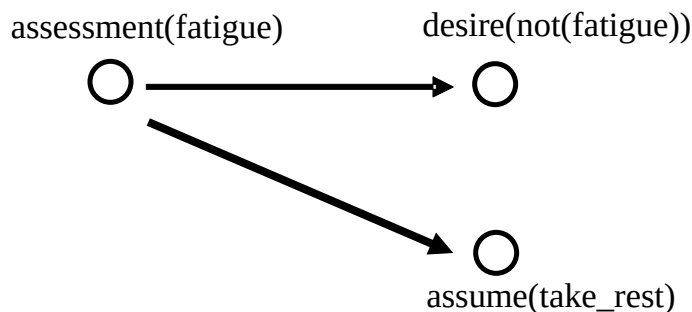
Examples:

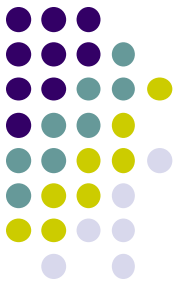
- `assumed(agent, performed(set_temp_to(21)))`
- `proposed(agent, performed(set_temp_to(21)))`



Additional Remark

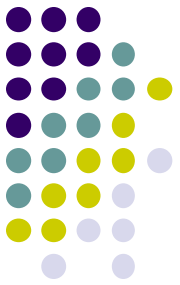
- **When** does a support model become active?
 - Once the analysis model has derived an **assessment**
- How can we **implement** this in a simulation?
 - Add rules from assessment to initial concepts support model:





Overview of today's lecture

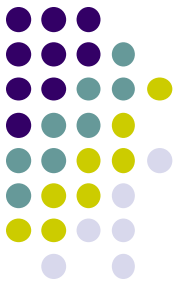
1. Summary previous lecture
2. Support models
3. **Ethical considerations**



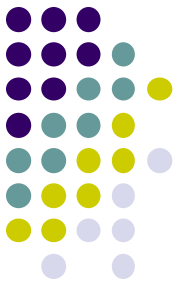
Ethical considerations

- This course aims to build intelligent systems that:
 - monitor people
 - reason about their mental/physical state
 - take initiative to support them
- This approach seems to have many **advantages**
- However: are there also any **downsides**?

Video: Smart support devices

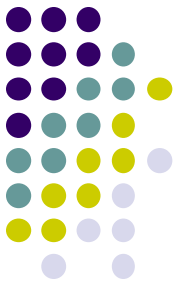


<https://vimeo.com/128873380>



Assignment

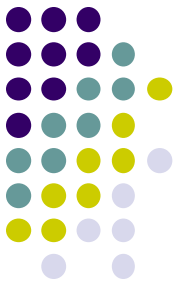
- Form groups of 3
- Write down at least 3 societal or ethical **concerns** of intelligent support systems
- List them one-by-one under “Discussion” in Canvas
 - Make sure your reasoning is understandable, so write sentences instead of key words



Next step

- Read them all and “like” the most relevant / serious ones
 - Individually
- Plenary discussion

Finally

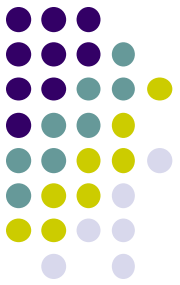


- How can we **deal with** these concerns?
 - Write down a possible solution / mitigation as “reply” to the most striking concerns
- Discussion



Questions to ask

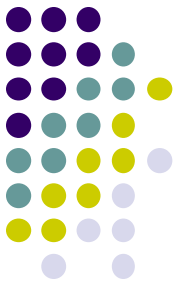
- What is at stake?
 - which persons?
 - which values?
- What are risks of:
 - humans being attached to all kinds of **sensors**?
 - systems having knowledge about our goals, emotions, intentions, ...? (**privacy** issues)
 - systems being **misused** by malicious parties?
 - systems **making autonomous decisions**?
 - systems **taking jobs**?
 - ...



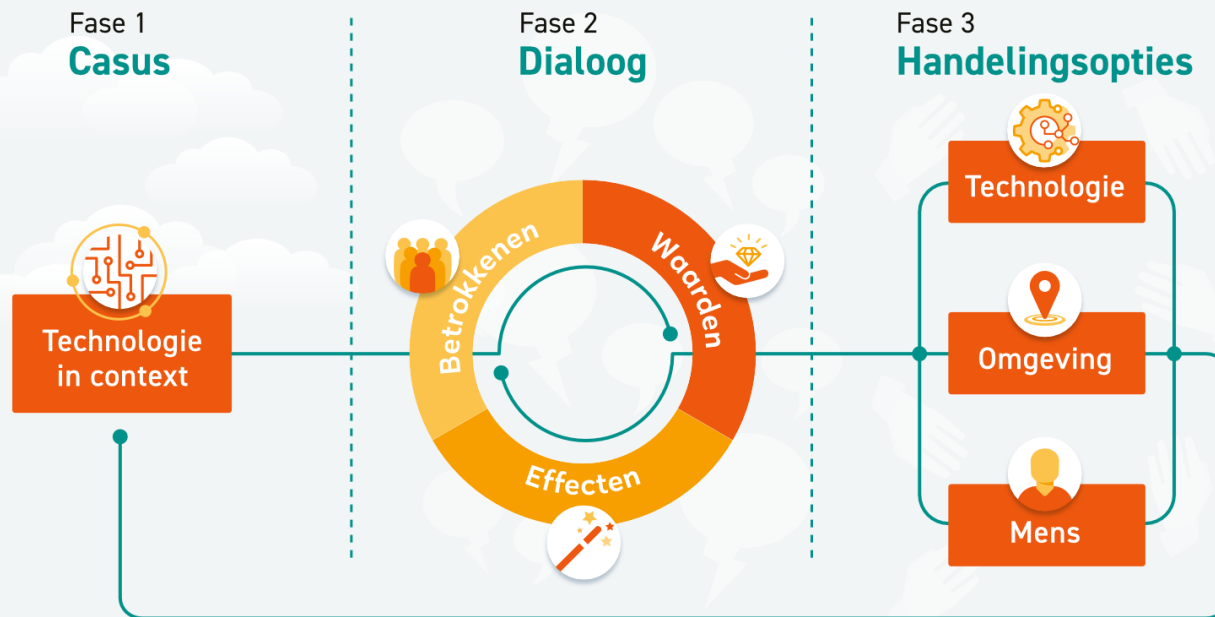
Possible Solutions

- **Transparency** of aims of system
- Increase users **awareness** of:
 - abilities of AI
 - (privacy) risks
- Adequate **legislation**
 - (e.g., explicit permission; ownership of data with users)
- Adequate **security**
- Explore various **modes** of system autonomy
- Extensive **evaluation** & testing
- Human in the loop
 - always build an **on/off button** 😊
- Discussions and consensus!

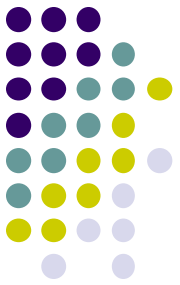
Approach for discussion



Aanpak begeleidingsethiek



<https://begeleidingsethiek.nl/>



Summary

- Intelligent support systems have aim of “understanding” and “support” people
- Serious concerns around privacy, autonomy, freedom
- As an AI expert, you need to be aware of this and have a responsibility
 - not only in application phase, but also in design phase: “**ethics-by-design**”